Continuous Caustic Collars and Billiard Rigidity

A Source Audit, Complex-Moment Refinements, and Ram-Represented Certificates

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Abstract

The manuscript under examination claims that a positively curved smooth planar billiard with a full continuous collar of smooth convex physical caustics is an ellipse. This is a global classification of tables, not a local perturbation theorem or a statement about a Cantor set of caustics. All eleven mathematical source sections and their references were read. We examine the physical-to-phase bridge, joint continuation, finite-endpoint algebraicity, critical moment laws, and the odd-periodic sign-test argument. Independently proved refinements replace infinite coalescent power-sum tests by sharp finite complex-moment tests, give quantitative root-cluster bounds, regularize division at vanishing kernel endpoints with a charged analytic error, quantify odd-cycle matrix inverses, and isolate a finite last-site probability certificate. Exact local Ram sources supply useful moment, representation, residual, and full-error interfaces; they do not supply the manuscript's global continuation or critical-wall exclusion. The report distinguishes rederived identities, conditional certificates, and unresolved verification obligations. It does not certify the complete rigidity proof or claim a new resolution of the Birkhoff conjecture.

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1 Target, Scope, and Assessment

1.1 The actual quantifiers

Let $\Omega \subset \mathbb{R}^2$ be bounded and strictly convex. Its boundary is a C^∞ simple closed regular curve with everywhere positive curvature. The physical collar in S1, `def:caustic-collar` is specified by a homeomorphism

$$H_0 : S^1 \times [0,1) \longrightarrow \mathcal{A} \subset \overline{\Omega}, \qquad H_0(S^1 \times \{0\}) = \partial\Omega,$$

where $\mathcal{A}$ is relatively open and contains the boundary. Every leaf with $0 < b < 1$ is a C^∞ simple closed regular caustic bounding a strictly convex set. The family need not be differentiable in b. Positive curvature of every inner leaf is not an additional stated hypothesis. Every intermediate leaf is required; a sequence or Cantor family accumulating at the boundary is insufficient.

The conclusion of S1, `thm:main` is that Ω is the interior of an ellipse, including a circle, after translation. There is no assumed central symmetry, closeness to an ellipse, algebraicity of a caustic, uniform bound on all derivatives, or foliation of the entire phase cylinder. These distinctions are necessary when comparing the claim with published partial results or with the private Ram report TR336.

1.2 What is established here

The L5 applicability of TR336 is justified by the same billiard map, graph residuals, periodic-action data, and rational cohomology operators. L5 measures direct relevance to the core problem. It does not mean that TR336 proves the classification. Its explicit warnings about rational rotation, finite truncation, and formal grazing data are particularly useful audit gates.

This report proves the finite theorems in Sections 4– 7 from their displayed hypotheses. Their main improvements are:

1. For a k-edge coalescent line model, the first k action coefficients suffice for isolation; fewer do not suffice over $\mathbb{C}$. For the point model, its $2k$ square-root terms reduce to k moment entries, and coefficients through order k again suffice.
2. Approximate complex moments give a root-radius enclosure through an explicit Newton majorant, not an unjustified positivity test.
3. Analytic endpoint division can be performed without dividing noisy values by two small endpoint fields. A projection enforces both vanishing axes and yields a uniform quotient error.
4. Odd-cycle invertibility admits explicit real margins and a conditional complex perturbation certificate. The small-field derivative suppression retains its stronger exponent.
5. A finite union of fixed-candidate stopped-martingale estimates gives an explicit last-site failure budget; no conditioning on a future-selected site is needed.

No counterexample to the manuscript's main theorem is established here. Conversely, reading the entire chain and finding coherent local mechanisms does not constitute an independent proof of all its difficult global lemmas. The highest-risk uncompleted verification tasks are identified in Section 9; they are not silently called demonstrated gaps. The local theorems below are mathematical refinements and certificate interfaces, not claims of historical priority.

2 Public Literature and Current Status

The following comparison was checked against public primary material on October 7, 2026. Only public titles and bibliographic metadata were used for external searches. External papers were checked at the stated theorem or abstract level; their complete proofs were not independently audited.

Kaloshin and Sorrentino's 2018 Main Theorem concerns domains close to a fixed ellipse, with C^k control for $k \geq 39$, a sufficiently small C^1 distance, and rational caustics of rotation $1/q$

for every $q > 2$ [2]. Thus its hypotheses include low-period caustics and perturbative closeness, neither of which follows merely by renaming a thin collar. Avila, De Simoi, and Kaloshin's earlier work treats small eccentricity [1]. These are not global classifications of arbitrary tables possessing only the near-boundary collar in S1.

Koval's Theorem 1 allows any fixed near-boundary rational cutoff and ellipses outside a locally finite exceptional set of eccentricities. It still requires C^k bounds, $k \geq 39$, and sufficiently small C^1 distance to the reference ellipse. The paper appeared online on December 11, 2025, in *Inventiones Mathematicae* 244 (2026), 221–298 [5]. It is a local strong Birkhoff theorem for almost every ellipse, not unrestricted collar rigidity. Its stated rational integrability includes all sufficiently small positive rational rotations.

Bialy and Mironov's result imposes central symmetry and a phase foliation extending to a four-periodic caustic [3]. Kaloshin, Koudjinan, and Zhang's Theorem 2 is perturbative near a centrally symmetric domain; it includes a three-periodic caustic and either four-rational integrability or the specified phase foliation [4]. Neither may be applied to an arbitrary collar lacking those periods and symmetry conditions. Global foliation of the whole phase cylinder is also a different condition: the cited Bialy theorem then gives a circle, not the whole class of elliptic tables.

Glutsyuk's 2026 theorem characterizes conics by existence of a complex *algebraic* caustic [6]. The algebraicity requirement is substantive; the local differential algebraicity of intermediate orbit data in S4 is not automatically such a caustic. Fierobe and Sorrentino study analytically parameterized families of twist maps: for a fixed rational rotation, periodic invariant curves occur at all parameters or at a discrete subset [7]. Their analytic-family assumption is not a shortcut from a merely continuous physical collar.

The primary material retrieved does not independently establish the unrestricted September 24, 2026 claim audited here. This is a bounded literature conclusion, not a claim that no later work exists. The pinned manuscript is treated as a source claim requiring its own complete proof; an older statement that the conjecture is open is not itself a refutation of that claim.

3 Reading the Source Chain

The identifiers S1–S11 refer to the exact files listed in Appendix A. Source labels are included so that the audit survives changes in PDF pagination.

3.1 Physical leaves, rational periodicity, and analyticity

For a positively oriented line with normal angle θ and offset p, the table support function h gives

$$S(\theta, \theta') = 2h\left(\frac{\theta + \theta'}{2}\right) \sin\left(\frac{\theta' - \theta}{2}\right), \qquad p = -S_1, \quad p' = S_2.$$

Writing $r = h + h'' > 0$ and $d = (\theta' - \theta)/2$ yields $S_{12} = r \sin d/2 > 0$ and $\partial\theta'/\partial p < 0$ on the chosen grazing branch. The sign convention matters: TR336's arc-length generating function must not be mixed with this oriented-line one without changing the associated momentum convention.

In S2, `lem:rational-periodicity`, nested physical leaves give continuous ordered support graphs filling a phase collar. Their induced circle maps are ordered by twist. Rational plateaus are excluded using area recurrence on an annulus: a rational circle map's nonperiodic points move monotonically under its period iterate and are nonrecurrent; almost-everywhere recurrence plus continuity and strict ordering then contradict a plateau. Nearby graphs with smaller and larger rotation force the rational iterate to be the prescribed translation everywhere. This is an actual argument addressing the warning in TR336, Section 13.1: rational rotation number alone does not imply that every point is periodic. The full filled collar, not one isolated leaf, is used in the upgrade.

S2 introduces the Lazutkin coordinate and $u = \log \phi'$ so that $r(\phi) = ce^{-3u}$. Its normalized antisymmetric action satisfies

$$D_1(X, X+T) = \frac{T^2}{8} + a(X)T^4 + O(T^5), \qquad a = \frac{7u'' - 8(u')^2 - e^{2u}}{384}.$$

The finite Dirichlet mesh is solved with a q-scaled norm; rational periodicity makes its critical action constant. The resulting high Fourier rows are close to the identity on a weighted Wiener space. The full high-mode operator, including off-diagonal rows and the omitted tail, is the object to invert. A finite matrix alone is not it.

For orientation, the common circle preconditioner can be written

$$DC_q(0)v = d_q v_q, \qquad d_q = \frac{q^3 \sin(\pi/q)}{\pi(q^2-1)}, \qquad d_q \in [d_N, 1) \quad (q \geq N).$$

On the corresponding high-mode Fourier–Wiener space the diagonal inverse norm is exactly d_N^{-1}. If a verified full operator error is $E < d_N$, the Neumann bound is $(d_N - E)^{-1}$. This last implication is immediate from factoring out the diagonal; the circle-row calculation and the source row estimates are separate inputs. Their detailed analytic-stage treatment is outside the principal scope of this report. In particular a reported estimate $E \leq C \|u\|_{r_0} / N$ must hold in the stated full operator norm, not just on a finite Fourier section. Appendix B records two additional analytic-stage refinements, checked after reading the independent local source review.

S3 does not infer convergence from an arbitrary formal grazing jet. It enforces rational value and first-parameter-jet compatibilities, divides two centered-difference operators on slanted complex domains, and uses a shrinking-domain iteration. Its conclusion is an even jointly analytic impact parametrization and a nonzero area coefficient $A(t) = ct/4 + O(t^3)$. Two rational compatibility conditions are needed for two divisions. The analytic continuation later in the manuscript depends on these joint germs, not solely on smooth or formal data.

3.2 Endpoint algebraicity and the first wall

In S4 the line family $L(y,t)$ and impact family $X(y,t)$ have maximal *joint meromorphic* widths. The relation

$$h_L(t) = h_X(t) + \frac{|\Im t|}{2}$$

uses intersections and reflection to transfer joint neighborhoods. It is not a statement about unrelated meromorphic slices.

The line lens theorem `thm:lens-lines` uses stationary action, finite support jets, Vandermonde elimination in distinct web directions, and a coalescing critical-value calculation to isolate the finite core. The point lens theorem `thm:lens-points` also requires the analytic area function $A(t)$ in its coefficient field. Endpoint positions alone are not its stated data. Noetherian finiteness does not give an explicit uniform endpoint jet order or algebraic degree. Section 4 below improves the finite coalescent model, not this missing effective description of the whole differential-algebraic elimination.

The meromorphic Hartogs lemma is stronger than an identity theorem in one variable: joint annular data and an open set of meromorphically filled slices produce a uniform finite-rank Laurent/Hankel relation locally in the parameter, then a denominator and a joint extension. The countable rank alternatives cannot simply be replaced by a fixed finite experimental matrix. The disk sweep uses coherent collars and an open seed, and the width comparison is superharmonic.

In S4, `thm:no-patch`, a hypothetical first-wall patch at a real parameter leads to nontrivial algebraic monodromy. An alternate branch descends to glancing, becomes even when transverse to the primary curve, and yields the operation A, I, A^{-1}, I, with $I : t \mapsto -t$. This replaces L_1 by L_{-1} while fixing a prescribed finite set of far endpoint jets. Shearing and repeating force too many distinct roots of one endpoint polynomial. A separate superharmonic-width argument

transfers generic-real-parameter information to the original parameter. Lower semicontinuity alone would not do so.

Two safeguards are explicit in the source and must be retained: the loop may depend on the chosen *finite* collection of far indices; it is not one loop fixing infinitely many fields. Also, a slice extension is promoted to a joint patch by recurrence in steep parameter sectors followed by bent analytic disks. That promotion is not automatic.

3.3 Shape classification and growth regimes

S5 uses the intrinsic identity

$$\frac{\det(\Gamma', \Gamma'')^4}{(\Gamma' \cdot \Gamma')^3} = c^2, \qquad a = e^{\mathrm{i}\phi}, \quad f = \phi', \quad \Gamma' = c e_\phi f^{-2}.$$

The scalar product is complex bilinear. Zeros or poles of a are isotropic centers. Their order is restricted by a rational glancing profile and ramification count; the surviving simple centers give reflections, an elliptic normal function, and $f^2 = C + B\cos 2\phi + D\sin 2\phi$. The support function of the associated positive definite quadratic form then recovers an ellipse up to translation. The complex-root and ramification arguments are not real Gram positivity arguments.

Without isotropic centers, the source constructs a single-valued holomorphic shape lift $\mathfrak{s}(y,t)$ into a strip D_H, despite possible poles of Γ. Its zeros of f are simple with $f'' = 0$ there, so $q = f''/f$ is holomorphic. In the mean-normalized gauge,

$$\mathfrak{s}(y,t) = y + t^2 V(y) + O(t^4), \qquad V' = \frac{f^2 + 8q - \langle f^2 + 8q \rangle}{120}. \tag{1}$$

The centering constant and the gauge cannot be dropped when comparing the physical strip width with the imaginary displacement.

S6 writes the fast fields $F = tf(s+tu)$ and $Q = t^2 q(s+tu)$, with $F'' = QF$, and three scalar kernels with potentials $Q_0 = Q$, $Q_\pm = Q \mp 3\mathrm{i}F'/2 - F^2/4$. Its cone estimate requires an exponent $\alpha > 1/2$, fixed aperture margins, a known lower slab, and an order of smallness choices. It excludes subcritical and finite superlinear growth by combining a quiet center with the global real width transfer. Pure limiting fields obey $FQ = 0$, not arbitrary coupled limiting dynamics.

S7 handles exponential profiles and infinite growth. The two frozen energies are

$$E_F = \frac{2(\sin z - z\cos z)}{\lambda^3}, \qquad E_Q = \frac{\tanh z - z\,\mathrm{sech}^2 z}{\lambda^3}, \qquad z = \lambda J/2.$$

Their first branch singularities, respectively a fractional-power and a logarithmic term, are tracked from the glancing germ along regular corridors. The precise branch, nonvanishing kernels, and nonzero twist along the corridor are part of the hypotheses. An endpoint value agreeing with a closed formula does not select a sheet. In the recurrence-to-ODE limit, the source subtracts the exact shifted identities before division by the mesh size, using analytic derivative convergence. It does not assume an unsupported $o(\text{mesh})$ rate from bare compact convergence.

3.4 Critical moments without a circular dependency

In S8 put $d_k = e^{-k}$ and

$$\mu_f(k) = d_k(\mathbb{E}|f|^2)^{1/2}, \quad \mu(k) = d_k(\mathbb{E}(|f|^2 + |q|))^{1/2}, \quad P = -\Im V.$$

An envelope has backward logarithmic rate $\beta < 1/2$ and a fixed forward rate. The one-sided estimate gives a negative-part cost b^m with $m < 2$, while a finite preliminary L^2 ratio C_* satisfies $C_*^2 \le C(1 + C_*^m)$. The strict exponent gap closes the bootstrap independently of the preliminary ratio. A concentration estimate improves the unweighted moment above two.

The ancestor-weighted bootstrap assumes its own coarse tile bounds and absolute ancestor P integral. It concludes weighted L^2 control, not the unweighted higher-moment improvement for free. Kernel comparability is used only after taking the signed harmonic integral under the actual harmonic-measure law. Replacing it prematurely by another measure would lose that equality.

S8 explicitly leaves its angle budget conditional on excluding the defined prohibited sequences. S9 proves that exclusion without using the conditional budget or polynomial bounds. The order is therefore not circular at the level of the displayed dependency chain. The limiting moment law is positive, log-convex in physical depth, normalized by $m(1) = 1$, and bounded by D^{-1}. Its contact dichotomy either gives a forbidden exponential tangent or a nonzero pure first field after re-centering and ancestor control.

For the latter field the limiting model has

$$J^3 g(z) = 1, \quad z = TF(S)J/2, \quad g(z) = \frac{3(\sin z - z\cos z)}{z^3}, \quad J = S_Y. \tag{2}$$

The deep anchor selects its branch. The twist derivative is $3J^2 \sin z/z$. For real parameters, the possible nonzero $n\pi$ degeneracies first form a countable set; continuity and Hurwitz on common connected interiors remove them, since the deep model tends to $z = 0$. Bent parameter graphs then give joint disks, not just real-parameter solutions. Their second coefficient satisfies $A'(Y) = F(Y)^2/120$, obtained from $g(z) = 1 - z^2/10 + O(z^4)$, and yields the upper growth bound.

The high-modulus mesh requires common relatively dense epochs and a witness near *every* horizontal center. Center-dependent subsequences would not suffice. Good chains are routed with bounded exceptional shadows and a subexponential total length. At a simple zero z_0, the condition F''/F holomorphic gives

$$F(z) = a(z - z_0) + O((z - z_0)^3), \qquad F(z)^{-2} = a^{-2}(z - z_0)^{-2} + O(1).$$

Thus F^{-2} has zero residues, a single-valued meromorphic primitive C_0, and $F(C_0 - c_*)$ is holomorphic across the zeros. One common constant c_* on the whole connected mesh is essential. The enclosed region estimates, reflection, and identity theorem then contradict $C_0' = F^{-2}$. This explains why mere large values on disconnected sets would not prove the source's conclusion.

3.5 Odd slice, history stops, and the sign row

S10 fixes one sufficiently large *odd* period N before the depth limit. Oddness makes both the alternating angle system and its offset system invertible. The real Jacobi matrix is a conjugated negative weighted cycle Laplacian, while its flat counterpart is positive definite. Proper truthful subsets and arbitrary flat subsets have nonzero determinants by a Schur complement.

The boundary-contact theorem first obtains almost-everywhere slice limits and then, if all sites stayed interior, promotes a regular slice orbit to a joint patch. Its return map is tangent to the identity with a shear, and steep return steps plus bent disks give the prohibited joint extension. Applying the no-patch theorem to a slice alone would omit this work.

Stopped geometry uses historical maxima, not current depths. Every fixed candidate modified matrix has its own estimates from the original start; a later choice among at most 3^N candidates does not restart a martingale. The source then improves a sublinear geometry exponent by $\sigma \mapsto (\sigma + \sigma^2)/2$, a finite number of iterations for each desired positive exponent. This does not assert one event simultaneously supporting every exponent.

S11 splits the sign-row proof into large and small moment regimes. In the large regime, stationary profile laws, exclusion of the zero profile, and complex WKB estimates make moderate $\log|f|$ values rare. The exceptional event where a nonzero limiting Q is real on an entire row is excluded by reflection and expected-modulus bounds. In the small regime, an amplitude variable $\log(|f| + \delta|f'|)$ avoids pointwise logarithmic singularities. A discrete Tanaka estimate

and curvature-witness occupation cost produce the deterministic row. The exponent arithmetic is consistent: with $\alpha = 1/100$, the two costs have powers $1/2 + 16\alpha = 0.66$ and $1 - 6\alpha = 0.94$, both less than one.

The robust sign disk removes small shadows of all nearby zeros, uses q to bound f'/f outside them, and then controls the entire disk. At the last site, the remaining intrinsic clock is estimated unconditionally from each first hit. Fixed candidate sets and clock cuts precede intersection with the terminal classification event. The final contradiction compares a classification probability at least $0.9 - o(1)$ with an $o(1)$ compulsory disk-exit probability. Section 7 makes this last comparison explicit.

4 Finite Complex Moments for Coalescent Isolation

4.1 The finite moment theorem and its stable form

The relevant moments are holomorphic power sums of complex entries, not squared moduli and not a positive semidefinite Gram form.

Theorem 4.1 (Finite complex support and a root-radius certificate)**.** *Let $\xi_1, \dots, \xi_k \in \mathbb{C}$, counting repetitions, and put $p_m = \sum_{i=1}^{k} \xi_i^m$. If $p_1 = \cdots = p_k = 0$, then all $\xi_i = 0$. More quantitatively, suppose $|p_m| \le \varepsilon_m$ for $1 \le m \le k$. Define*

$$E_0 = 1, \qquad E_j = \frac{1}{j} \sum_{m=1}^{j} E_{j-m} \varepsilon_m \quad (1 \le j \le k). \tag{3}$$

If $R > 0$ and $\sum_{j=1}^{k} E_j R^{-j} < 1$, then $|\xi_i| < R$ for every i. For $k \ge 2$, the first $k-1$ zero moments alone do not imply the exact conclusion.

Proof. Write $\prod_i (z - \xi_i) = \sum_{j=0}^{k} (-1)^j e_j z^{k-j}$, where $e_0 = 1$. Newton's identities give

$$j e_j = \sum_{m=1}^{j} (-1)^{m-1} e_{j-m} p_m.$$

Induction proves $|e_j| \le E_j$. If all the power sums vanish, all e_j vanish and the polynomial is z^k. In the approximate case, a root with modulus at least R would satisfy $1 \le \sum_{j=1}^{k} |e_j| |z|^{-j} < 1$, a contradiction. Finally, take $\xi_i = a\omega^{i-1}$, $a \ne 0$, where $\omega = e^{2\pi \mathrm{i}/k}$. Then $p_m = 0$ for $1 \le m < k$ but $p_k = k a^k \ne 0$. □

The last family also shows why a kth-root scale in an approximate cluster bound is unavoidable: an error of size $|a|^k$ in the final moment can hide nodes of size $|a|$. Approximate zeros cannot certify exact isolation without a separate exact-vanishing argument. Multiplicity and the correct support count are part of the data.

4.2 Line endpoints: a sharp finite action-jet test

Theorem 4.2 (Line coalescent model)**.** *Let $k \ge 2$, $r \in \mathbb{C} \setminus \{0\}$, and $f_1, \dots, f_k \in \mathbb{C}$. Near $x = 0$, take the analytic critical branch with $z_i(0) = 1/k$ of*

$$G(x) = \operatorname{crit}_{\sum z_i = 1} \sum_{i=1}^{k} \left(-\frac{r}{24} z_i^3 + x f_i z_i \right), \qquad G_0 = -\frac{r}{24k^2}. \tag{4}$$

Then

$$G(x) - G_0 = O(x^{k+1}) \quad \Longleftrightarrow \quad f_1 = \cdots = f_k = 0.$$

The order k cannot be lowered uniformly over complex f_i.

Proof. The constrained Hessian at zero is $-r/(4k)$ times the bilinear Euclidean form on $\sum \delta z_i = 0$, which is nondegenerate. Thus the implicit function theorem supplies the unique local branch. The stationary equations and Euler identity are

$$-\frac{r}{8}z_i^2 + xf_i = \Lambda, \qquad G' = \sum_i f_i z_i, \qquad \Lambda = 3G - 2xG'. \tag{5}$$

If $G - G_0 = O(x^{k+1})$, then $\Lambda = -r/(8k^2) + O(x^{k+1})$. With the square-root branches equal to one at zero, this implies

$$z_i = \frac{1}{k}\sqrt{1 + \frac{8k^2}{r}xf_i} + O(x^{k+1}).$$

Summing and comparing coefficients of x^m, $1 \le m \le k$, gives $\sum_i f_i^m = 0$: each binomial coefficient $\binom{1/2}{m}$ is nonzero. Theorem 4.1 finishes this direction; the converse is immediate.

For sharpness, use the nonzero root-of-unity list in that theorem. Permuting the variables in (4) shows $G(\omega x) = G(x)$ by uniqueness of the critical branch. Hence its coefficients of orders $1, \dots, k-1$ vanish although the list is nonzero. The already proved implication forces the coefficient of order k to be nonzero. □

This is the exact model obtained in S4, `thm:lens-lines` after coalescing endpoints and scaling the first nonzero deformation. The source invokes every positive power sum; only the first k are needed for this model. For real f_i the second power sum would already be decisive, but the endpoint algebraic argument is complex. Substituting a real-square argument there would be invalid.

4.3 Point endpoints: removing repeated moment entries

Theorem 4.3 (Point coalescent model)**.** *Let $k \ge 2$, $r \ne 0$, $f_0 = f_k = 0$, and $f_1, \dots, f_{k-1} \in \mathbb{C}$. On the analytic critical branch $z_i(0) = 1/k$, $\sum z_i = 1$, set*

$$G(x) = \operatorname{crit} \sum_{i=1}^{k} \Big[-\frac{r^2 z_i^3}{24} - \frac{rx(f_i + f_{i-1})z_i}{2} + \frac{x^2(f_i - f_{i-1})^2}{2z_i} \Big], \qquad G_0 = -\frac{r^2}{24k^2}. \tag{6}$$

If $G = G_0 + g_1 x + O(x^{k+1})$, then $g_1 = 0$ and every $f_i = 0$. Conversely the zero list has this property.

Proof. The Hessian at zero is again nondegenerate on the constraint hyperplane, now with coefficient $-r^2/(4k)$. The multiplier obeys $\Lambda = 3G - 2xG'$ because every summand has weighted degree three when z has weight one and x has weight two. Directly differentiating the displayed expression verifies this identity, including its last inverse-z_i term. Stationarity also gives

$$\begin{aligned} \Big(\frac{rz_i}{2} + \frac{x(f_i - f_{i-1})}{z_i}\Big)^2 &= -2\Lambda - 2rxf_{i-1}, \\ \Big(\frac{rz_i}{2} - \frac{x(f_i - f_{i-1})}{z_i}\Big)^2 &= -2\Lambda - 2rxf_i. \end{aligned} \tag{7}$$

Here $\Lambda = 3G_0 + g_1 x + O(x^{k+1})$. Both roots start at $r/(2k)$; adding them over all edges and using $\sum z_i = 1$ therefore yields, with $\beta = 8k^2/r^2$,

$$2\sqrt{1 - \beta x g_1} + 2\sum_{i=1}^{k-1} \sqrt{1 - \beta x(g_1 + rf_i)} = 2k + O(x^{k+1}).$$

The two endpoint entries coincide because $f_0 = f_k = 0$; each interior entry also occurs twice. Divide by two before extracting moments. The k entries $g_1, g_1 + rf_1, \dots, g_1 + rf_{k-1}$ have their first k power sums zero. Theorem 4.1 makes them all zero; the entry g_1 and $r \ne 0$ give the conclusion. □

This applies to the limiting point model in S4, `thm:lens-points`. It does not discard the area coefficient from the original endpoint field. Nor does it prove that a prescribed finite set of unscaled boundary jets isolates the entire original analytic core. Passing from the first nonzero deformation to this model remains the source's formal elimination and coalescence argument.

4.4 The precise Ram moment/operator counterpart

For a finite list ξ, introduce the complex atomic evaluation functional

$$\mathcal{L}_\xi(p) = \sum_{i=1}^{k} p(\xi_i).$$

Its first k monomial moments determine the monic support polynomial *with multiplicities* by Newton identities. This is a concrete moment representation of the source's coalescent algebra, not a new inversion of arbitrary analytic functions. The quotient algebra $\mathbb{C}[z]/(\prod_i (z-\xi_i))$ gives a k-dimensional multiplication operator; its characteristic polynomial is the same monic polynomial. Repeated nodes can give nonsemisimple multiplication and are not handled by assuming a nonsingular distinct-node Vandermonde matrix.

TR247's `thm:moment-inertia` and `eq:moment-pencil` concern distinct conjugation-stable nodes with matched positive weights. Under those extra hypotheses they give a bilinear Hankel inertia test and exact support pencil. The present f_i need not satisfy those hypotheses. For example the list $(\mathrm{i}, -\mathrm{i})$ has $p_2 = -2$ despite positive atomic weights. A Hermitian evaluation Gram would be positive but would encode different mixed moments; its positivity cannot establish the needed holomorphic power-sum vanishing. TR2405's contour-count/Newton construction provides the appropriate finite multiplicity-aware interface when the entries are known only through a certified analytic zero count.

5 Branch-Safe Continuation and Kernel Division

5.1 Finite monodromy obstruction and its limits

Proposition 5.1 (A finite root count suffices)**.** *Let K be a field of meromorphic germs of characteristic zero and $P \in K[Z]$ a nonzero polynomial of degree D. Suppose a specified continuation construction fixes all coefficients of P and produces $D+1$ pairwise distinct root germs. These assumptions are contradictory. It suffices to construct a finite collection of continuation operations, each fixing the finitely many coefficient germs needed at its step.*

Proof. Embed all the finitely many resulting algebraic germs in a common algebraic extension of K. Successive division by their distinct linear factors would make a degree D polynomial divisible by a degree $D+1$ polynomial. No common operation on an infinite collection is required for this argument. □

For the source's branches $L_1, L_{-1}, \ldots, L_{1-2D}$, a generic projective coordinate has different first t derivatives at glancing, proportional to the distinct indices and a nonzero ϕ' factor. This proves distinctness as germs. A finite numerical separation would additionally need enclosed higher derivatives and a verified degree D; neither is supplied by the abstract finite-root argument. The source's Noetherian step is qualitative. This proposition clarifies its finite logical content without converting it into an effective algorithm for all billiard tables.

Remark 5.2 (Finite contour moments do not fill a disk). For any $m \geq 1$, the annular holomorphic function $A(\zeta) = \zeta^{-(m+1)}$ satisfies $\oint \zeta^j A(\zeta)\,\mathrm{d}\zeta = 0$ for $0 \leq j < m$ but has a pole at the center. Thus finitely many vanishing Cauchy moments do not replace the all-moment identity in S9's `model:graph-filling`. A meromorphic finite-rank certificate needs a proved denominator/rank bound and the complete recurrence, not a sampled Hankel minor. This is a counterexample to a tempting finite upgrade, not to the source's all-moment proof.

5.2 An analytic quotient with exact vanishing axes

The kernel action in S6, `lem:kernel-division` has numerator

$$N = k_0(J) - \sqrt{k_+(J)k_-(J)}, \qquad p = \frac{N}{F(0)F(J)}.$$

The square root is continued from glancing. On a panel where $k_0(J) \neq 0$, initial data $(F(0), F'(0))$ can be replaced locally by endpoint coordinates $(A, B) = (F(0), F(J))$; the Jacobian determinant is $k_0(J)$. The source's identities give $N(A, 0) = N(0, B) = 0$. The following bound retains that exact structure.

Theorem 5.3 (Stable double division)**.** *Suppose N is holomorphic on the bidisk $|A| < R_A$, $|B| < R_B$, with $|N| \leq M$ and zero on both coordinate axes. Then $Q = N/(AB)$ extends holomorphically and*

$$|Q| \leq \frac{M}{R_A R_B}. \tag{8}$$

Writing $Q = \sum_{i,j\geq 0} q_{ij} A^i B^j$, one has $|q_{ij}| \leq M R_A^{-i-1} R_B^{-j-1}$. On $|A| \leq \theta_A R_A$, $|B| \leq \theta_B R_B$, with $0 \leq \theta_A, \theta_B < 1$, its rectangular truncation through degrees (n, m) has error at most

$$\frac{M}{R_A R_B} \frac{\theta_A^{n+1} + \theta_B^{m+1} - \theta_A^{n+1}\theta_B^{m+1}}{(1-\theta_A)(1-\theta_B)}. \tag{9}$$

If a holomorphic approximation $\widetilde{N}$ has uniform error at most ε on the bidisk, set

$$\widetilde{N}_c(A, B) = \widetilde{N}(A, B) - \widetilde{N}(A, 0) - \widetilde{N}(0, B) + \widetilde{N}(0, 0).$$

Then $\widetilde{Q}_c = \widetilde{N}_c/(AB)$ is holomorphic and $\left\|Q - \widetilde{Q}_c\right\|_\infty \leq 4\varepsilon/(R_A R_B)$.

Proof. The power series of N has no term on either coordinate axis, so division is removable. Apply Schwarz's lemma in A with B fixed to get $|N/A| \leq M/R_A$. This quotient vanishes at $B = 0$, so Schwarz in B gives (8). Cauchy's coefficient estimates, first on smaller radii and then by passage to their suprema, give the coefficient bound. Sum the geometric series over $i > n$ or $j > m$, subtracting the double-counted corner, to obtain (9). The corrected numerator vanishes on both axes and differs from N by at most 4ε. Apply the same double-division bound to that difference. □

There is also the exact anchored integral

$$Q(A, B) = \int_0^1 \int_0^1 \partial_A \partial_B N(sA, tB)\, \mathrm{d}s\, \mathrm{d}t,$$

including $A = 0$ or $B = 0$. A tensor polynomial approximation to this holomorphic quotient can be stored in the unisolvent moment coordinates of TR250, `eq:moments`, or directly in Taylor coefficients. The representation error is still (9) plus the charged numerator error; changing coordinates does not remove it.

The hypotheses require a complex-neighborhood supremum, a coherent root, and a valid endpoint chart. Samples on a real mesh do not bound the holomorphic error. The theorem does not cross $k_0(J) = 0$, a square-root branch collision, or the first shape wall. Such points need another chart or the source's separate analytic argument.

5.3 A quantitative regular-corridor tracking gate

Proposition 5.4 (Charged tracking on a fixed corridor)**.** *Let $z(t)$ solve $z' = V(z)$ for $0 \le t \le T$ in a fixed tube. Assume V is Lipschitz there with constant L, the solution has distance at least $r > 0$ from the tube boundary, and a computed recurrence on a mesh of step $h > 0$ satisfies*

$$\|z_{j+1} - z_j - hV(z_j)\| \le h\delta + Ch^2.$$

Suppose the exact solution has one-step Euler defect at most C_0h^2 and $jh \le T$. Then, while the iterates stay in the tube,

$$\|z_j - z(jh)\| \le e^{LT}\big(\|z_0 - z(0)\| + T[\delta + (C + C_0)h]\big). \tag{10}$$

If the right side is less than r, the same estimate closes the tube condition inductively. The defect is charged over the whole computed interval, not just a portion shared with another trajectory.

Proof. Subtract the exact Euler step. With $e_j = \|z_j - z(jh)\|$, $e_{j+1} \le (1+Lh)e_j + h\delta + (C+C_0)h^2$. Iterate and use $(1 + Lh)^j \le e^{LT}$ and $jh \le T$. The strict boundary margin gives induction on the first possible exit step. A shorter final step is treated with its actual length. □

For S7 and S9 the state is (S, J), and the limiting vector field is

$$\left(J, \frac{p_S - Jp_{SJ}}{p_{JJ}}\right).$$

The proposition is applicable only after enclosing p_{JJ} away from zero and deriving the displayed local defect on the selected sheet from the exact stationary recurrence. The action quotient certificate above can contribute to that enclosure. Uniform closeness of actions without derivative bounds is not enough. Nor is a per-step $O(\delta)$ error enough: its accumulated cost would be $O(\delta/h)$, unlike the charged $h\delta$ here. If two continued model germs are separated by a certified distance $s_* > 0$, two such tracking bounds each below $s_*/2$ can distinguish them. Producing the actual global single-valued domain in which they would have to agree remains a separate source obligation.

6 Odd-Cycle Operators and Quantitative Margins

Fix an odd $N \ge 3$. Let P be cyclic shift on $\mathbb{C}^N$ and $C_N = I + P$. For positive real edge lengths ℓ_j, define

$$M_0 = C_N^{\mathsf{T}} \operatorname{diag}(\ell_j^{-1}) C_N.$$

This is exactly the flat matrix in S10, `odd:Jacobi-matrix`, not a surrogate Gram matrix of unrelated features.

Theorem 6.1 (Exact odd inverse and a real spectral margin)**.** *For odd N,*

$$C_N^{-1} = \frac{1}{2}\sum_{m=0}^{N-1}(-P)^m, \qquad \left\|C_N^{-1}\right\|_\infty = \frac{N}{2}, \qquad \sigma_{\min}(C_N) = 2\sin\frac{\pi}{2N}.$$

Consequently

$$M_0 \succeq \frac{4\sin^2(\pi/(2N))}{\ell_{\max}} I. \tag{11}$$

For even N, C_N has the alternating null vector, so neither invertibility conclusion holds.

Proof. Multiply the finite geometric sum by $I+P$ and use $P^N = I$: the product is $I-(-P)^N = 2I$. Each row of the inverse has N entries of modulus $1/2$, giving the induced infinity norm and its attainability by aligned phases. The unitary discrete Fourier basis gives eigenvalues $1 + e^{2\pi i m/N}$; their minimum modulus for odd N is $2\sin(\pi/(2N))$. Finally $x^* M_0 x = \sum_j |(C_N x)_j|^2/\ell_j$ on the real positive-length slice, which proves (11). The even alternating vector is immediate. □

Theorem 6.2 (Uniform real modified-block inverse)**.** *Assume real $w_j > 0$, $\ell_j > 0$, and a real symmetric matrix $\mathcal{H} = M_0 - \mathcal{V}$ with $\mathcal{V}$ diagonal and $\mathcal{H}w = 0$. Put*

$$\gamma_j = \frac{w_j w_{j+1}}{\ell_j}, \quad a_* = \frac{4\gamma_{\min}\sin^2(\pi/(2N))}{w_{\max}^2}, \quad b_* = \frac{4\sin^2(\pi/(2N))}{\ell_{\max}}.$$

For a proper truthful set I and a flat set $F \subseteq I^c$, let $B = B_{I,F}$ be the source's retained matrix. If both blocks are nonempty and $D = (M_0)_{F,F}$, $C = (M_0)_{I,F}$, then

$$\left\|B^{-1}\right\|_2 \le \left(1 + \frac{\|C\|_2}{b_*}\right)^2 \max(a_*^{-1}, b_*^{-1}). \tag{12}$$

If only the truthful block is present the bound is a_*^{-1}; if only the flat block is present it is b_*^{-1}. The empty matrix requires no inverse estimate.

Proof. Conjugation by $D_w = \mathrm{diag}(w_j)$ gives a negative weighted cycle Laplacian: its off-diagonal entries are γ_j and its row sums vanish. Extend any vector on the proper set I by zero at the other sites. Deleting at least one vertex cuts the cycle into Dirichlet paths of at most $N-1$ interior vertices. The unweighted Dirichlet path with m interior vertices has least eigenvalue $4\sin^2(\pi/(2(m+1)))$, as verified by its sine eigenvectors. Thus $-\mathcal{H}_{I,I} \succeq a_* I$ after undoing D_w. Every principal flat block has the margin b_* from (11). With $A = \mathcal{H}_{I,I}$,

$$B = \begin{pmatrix} I & CD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} A - CD^{-1}C^{\mathsf{T}} & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} I & 0 \\ D^{-1}C^{\mathsf{T}} & I \end{pmatrix}.$$

The Schur complement is at most $-a_* I$. Each inverse triangular factor has norm at most $1 + \|C\|_2 / b_*$. Taking norms proves (12). The single-block cases follow from their respective margins. □

The hypothesis $\mathcal{H}w = 0$ fixes the diagonal used in the proof; positive off-diagonal entries alone do not fix it. One may bound $\|C\|_2 \le 4/\ell_{\min}$ if a coarser fully explicit estimate is preferred. These constants quantify the real-collar starting determinant margins in S10. They do not remain positive by complex continuation: away from the real slice, the matrices are complex symmetric, not Hermitian.

Corollary 6.3 (Complex panel and residual gate)**.** *Let B_0 be any invertible reference matrix and suppose on a complex panel $\left\|B_0^{-1}(B - B_0)\right\| \le \vartheta < 1$ in a fixed induced norm. Then*

$$\left\|B^{-1}\right\| \le \frac{\left\|B_0^{-1}\right\|}{1 - \vartheta}.$$

If a computed vector $\widehat{z}$ has residual $\|B\widehat{z} - b\| \le \eta$, then $\|\widehat{z} - B^{-1}b\| \le \left\|B_0^{-1}\right\| \eta/(1-\vartheta)$.

Proof. Factor $B = B_0[I + B_0^{-1}(B - B_0)]$ and sum the norm-convergent Neumann series. Apply the inverse to the residual. □

In moment coordinates, choose invertible analysis maps E for the retained physical vectors and represent the *same* linear map by EBE^{-1}, with norm transported by E^{-1}. The bounds then have identical content. Using the raw Euclidean norm of badly scaled moment coordinates instead changes the constants by their condition numbers. TR336's full-space block warning is respected here: this is a genuinely finite fixed odd slice, not an inverse on all Fourier modes of the billiard.

7 A Finite Last-Site Certificate

7.1 Retaining the two different suppression exponents

Proposition 7.1 (Exact two-channel suppression)**.** *At a nondegenerate site suppose the exact source identities hold:*

$$w_i = c\sin d_i\, f_i^{-2} s_i', \qquad 2f_i s_i' = (M_0 w)_i, \qquad c\sin d_i \neq 0, \quad f_i \neq 0.$$

Write $a_i = \log|f_i|$ *and* $W = \max_j |w_j|$*. Then*

$$|s_i'| \le \min\left\{ \frac{e^{2a_i}}{|c\sin d_i|}, \frac{\sum_j |(M_0)_{ij}|}{2} e^{-a_i} \right\} W. \tag{13}$$

If $|c\sin d_i|^{-1} \le A$ *and the half row sum is at most* B*, then on* $a_i \le -h$ *the first bound is* $Ae^{-2h}W$*, whereas on* $a_i \ge h$ *the second is* $Be^{-h}W$*.*

Proof. Solve the first identity for s_i' and take its modulus. Apply the row-sum norm to the second identity. Their minimum and the two stated sign restrictions give the result. □

The source uses the weaker symmetric bound $e^{o(h_2)-|a_i|}W$ after explicitly deriving both channels. Keeping the first exponent $2h$ can improve a finite certificate on a known negative-sign disk. This does not improve the source's global asymptotic sign-row theorem: the selected last site's sign may be positive, and either sign must be allowed.

7.2 Stopping before selecting a terminal classification

Theorem 7.2 (Finite fixed-candidate exit budget)**.** *Let* $\mathcal{A}$ *be a finite set of candidates. For each* $a \in \mathcal{A}$ *let* $T_a \le \tau$ *be a stopping time and* Z^a *a continuous complex local martingale with* $Z^a_{T_a} = 0$*. Its real and imaginary parts have quadratic variations* Q^a *and zero mutual covariation. Let* η_a *be first exit from the disk of deterministic radius* $r_a > 0$*, capped at* τ*. Suppose an event* G *is partitioned or covered by candidate events* G_a *(not required to be adapted) such that on* G_a*: disk exit is compulsory and* $Q^a_{\eta_a} - Q^a_{T_a} \le q_a$*, where* $q_a > 0$ *is deterministic. Then*

$$\mathbb{P}(G) \le 4 \sum_{a\in\mathcal{A}} \frac{q_a}{r_a^2}. \tag{14}$$

More generally, if the asserted properties hold outside a failure event of probability at most δ*, add* δ *to the right side. No martingale estimate conditional on* G_a *is asserted or needed.*

Proof. For each fixed a, stop also when its scalar clock after T_a reaches $2q_a$. Localization and the two quadratic-variation identities give expected squared stopped displacement at most $4q_a$. The nonnegative submartingale maximal inequality, or optional stopping at first disk exit, bounds the probability of reaching r_a before that cut by $4q_a/r_a^2$. On G_a, the extra cut cannot precede η_a, so G_a is contained in this exit event. Take the finite union; finally remove the failure event in the more general case. Unbounded terminal times are handled by localization and monotone convergence of the exit events. □

For S11 take $a = (J, i)$ with nonempty $J \subseteq \{0, \dots, N-1\}$ and $i \in J$, so there are $N2^{N-1}$ candidates. Define $T_J = \max_{j\in J}(\sigma_j \wedge \tau_L)$ before classifying the terminal path. Each finite maximum is a stopping time. The source yields a radius and clock cap of the form

$$r = de^{-h_1/4}, \qquad q = d^2 e^{Ch_2 - 2h_1}, \qquad h_2 = K^{\alpha/4}, \quad h_1 = \tfrac{1}{2}K^{\alpha/2}.$$

Consequently its compulsory-exit classification probability is at most

$$\delta_{\text{geom}} + \delta_{\text{inv}} + \delta_{\text{sign}} + \delta_{\text{clock}} + 4N2^{N-1}\exp(Ch_2 - \tfrac{3}{2}h_1). \tag{15}$$

The source proves these δ terms tend to zero for its fixed odd slice, and a history-gap classification has probability greater than 0.9. Hence its asymptotic contradiction follows *conditional on those preceding source theorems.* Formula (15) identifies what an explicit finite-K verification would additionally need: numerical bounds for every failure term and for C, not just a small final exponential.

On a verified negative-sign disk, Proposition 7.1 instead gives a clock factor e^{-4h_1}, so with the same radius its corresponding exit term has $e^{Ch_2-7h_1/2}$. The positive-sign term remains $e^{Ch_2-3h_1/2}$. The improvement is local and sign-sensitive, not a claim that every last site belongs to the better case.

7.3 A checkable robust sign panel

Proposition 7.3 (Error budget for a sign disk)**.** *Suppose f is holomorphic and nonzero on a disk $\overline{D(z_0,r)}$ and $|f'/f| \le L$ there. A certified enclosure $\log|f(z_0)| \in [a_-, a_+]$ implies*

$$\log|f(z)| \in [a_- - Lr,\ a_+ + Lr] \quad (|z - z_0| \le r).$$

Thus $a_- - Lr \ge h > 0$ certifies the positive channel throughout the disk, while $a_+ + Lr \le -h$ certifies the negative channel.

Proof. A nonvanishing holomorphic function on a disk has a holomorphic logarithm. Integrate f'/f along the segment from z_0 to z, then take real parts and the enclosure endpoints. □

Zero-freeness is a hypothesis, not a consequence of a positive sample. It can be established by a full contour margin and argument-principle count, or by the ODE/Rouche zero test used in S11, `prop:robust-sign`. If an approximation $\widehat{f}$ is used, one must charge both field and derivative errors. For example, on a whole disk, $|\widehat{f}| \ge m > \varepsilon_0 \ge |f - \widehat{f}|$ and $|\widehat{f'}| \le M_1$, $|f' - \widehat{f'}| \le \varepsilon_1$ imply $|f'/f| \le (M_1 + \varepsilon_1)/(m - \varepsilon_0)$. These are neighborhood bounds, not finite unvalidated samples.

8 Ram Applicability Without Claim Inflation

TR336 is unusually close to the actual problem. Its support-function representation keeps the curvature condition $h + h'' > 0$, translation gauge, and geometric map. Its graph residual

$$\mathcal{F}(\gamma)(s) = P(s, \gamma(s)) - \gamma(S(s, \gamma(s)))$$

uses the actual billiard map (S, P), not an arbitrary transform. Its `thm:exclusion` is a nonzero-residual exclusion on a declared C^0 neighborhood; its derivative, quadratic remainder, mesh interpolation and map-error budgets specify what is being excluded. Its rational rotation inverse includes the full cyclic cokernel. Its periodic-action theorem and uniform fixed-polygon remainder provide useful necessary conditions near the circle. None of these alone upgrades isolated invariant graphs to a full caustic collar.

The contribution here uses that same discipline in the more difficult complex stage. The actual finite moment functional is $\mathcal{L}_\xi$ from Section 4; the actual finite matrix is the retained odd Jacobi system; the actual analytic quotient is the kernel action; the actual observed quantities are sign margins and stopped quadratic clocks. These concrete objects can be represented in Ram coordinates without changing their branches, physical norm, or probability law.

TR250 provides the exact finite map $m = Ec$ with a specified unisolvent bank. TR247 supplies a finite support/Hankel theorem under stronger node hypotheses than the complex coalescent model. TR2405 supplies argument-principle counts and Newton recurrences only with a nonzero contour margin, validated quadrature, a unique integer count, and complete multiplicity enclosures. These sources were read at their actual local theorem locations, not inferred from catalog summaries. No unrelated number-theoretic conclusions are transferred to billiards.

There are several genuine benefits: the finite moment test makes the coalescent isolation order transparent; the quotient projection avoids a small-denominator numerical artifact; the explicit odd matrix margin gives a starting scale for complex panels; and (15) distinguishes each probabilistic failure cost. There are also costs. Moment bases can be ill conditioned, multiple roots are intrinsically unstable, finite panels require complete coverage and tails, and the candidate matrix count grows exponentially with the fixed period. None of these representations reduces the core global continuation burden merely by changing notation.

9 Verification Boundaries and Research Directions

9.1 Demonstrable limitations versus unchecked steps

The following distinctions are part of the mathematical conclusion.

Independently proved here.
The finite complex-moment and two coalescent theorems, their root majorant, double-division estimate, finite root-count observation, corridor error propagation, odd-cycle and modified-block bounds, two-channel suppression, and stopped exit budget have complete proofs above. They require exactly their displayed hypotheses.

Checked source identities and logical ordering.
The generating-function signs; the Euler multipliers and point square-root identities; the 1/120 model coefficient; the odd cycle Laplacian and cofactor mechanism; the exponent arithmetic in the sign occupation estimate; and the finite-candidate versus terminal-event distinction were rederived or traced explicitly. The conditional angle budget is not used to prove its own sequence-exclusion input in the displayed source chain.

Not independently completed.
The entire finite-core differential-algebraic isolation and specialization proof, global monodromy construction, every uniform cone/diagonal selection, every parameter-disk gluing step, and the complete high-modulus mesh and stopped-geometry probability chain have not been rebuilt into an independent proof of the original rigidity theorem. Reading and checking their main derivations is not full formal or peer certification. The analytic inverse and rational-resonance Newton stages also retain their source estimates as obligations.

Demonstrably invalid upgrades.
Finitely many annular moments do not give holomorphic filling; $k-1$ complex power sums do not isolate k nodes; real positive definiteness does not persist as Hermitian positivity on complex panels; a rational rotation number alone is not full periodicity; and a C^∞ flat grazing residual does not imply an exact analytic band. The counterexamples or hypotheses are given above or in the directly read TR336. These warn against shortcuts and are not asserted to be errors in the source's stronger arguments.

No demonstrable contradiction in the source's main theorem was found in this bounded audit. That is deliberately weaker than certifying the theorem. The source contains substantive proofs addressing many of the known gates; it would be inaccurate to describe them all as merely missing assumptions. It would also be inaccurate to regard the finite refinements in this report as substitutes for all of them.

9.2 Concrete next mathematical tasks

First, an effective version of the finite-endpoint algebraicity argument could seek bounds on core dimension, elimination order, and polynomial degree for a fixed number of impacts. The sharp coalescent tests reduce one part to finitely many moments, but do not yet supply those bounds. A result uniform over a geometric class would need explicit curvature, analyticity-width, and nondegeneracy assumptions.

Second, a branch-certified implementation could cover a chosen regular model corridor by overlapping complex panels, store endpoint-regularized kernel quotients, and maintain interval bounds for twist, branch separation, and local recurrence defects. A successful finite run would certify that corridor, not the absence of all possible shape walls. Failure at a degenerate panel is inconclusive unless an analytic alternative is proved.

Third, the odd-cycle estimates suggest a quantitative finite-K version of the last-site argument for a specified table and slice. It must bound every term in (15), including the exceptional sign-row fraction and the geometry and determinant failures. The stronger negative-sign exponent may reduce the required scale in favorable panels. The source's asymptotic diagonal arguments currently provide no numerical universal threshold through this report.

Finally, a stability theorem for approximately foliated tables would need a new bridge: errors in periodic actions, collar gaps, and analytic continuation must imply a controlled distance to an ellipse in a specified norm. The local root-radius and residual bounds are plausible ingredients, but the exact rigidity argument is not automatically stable under such perturbations. No quantitative near-ellipse theorem is claimed here.

10 Conclusion

The manuscript's target is the full physical-collar classification for arbitrary positively curved smooth planar tables. The exact Ram corpus is directly applicable to its billiard residuals, periodic actions, finite moments, and operator certificates. Its applicability does not by itself settle the global theorem.

The principal derived gains are sharp finite complex-moment tests for the line coalescent model, a reduced finite moment test for the point model, stable analytic kernel division, explicit odd-cycle inverse margins, and a sign-sensitive finite last-site error budget. Their proofs and limits are separate from the source's difficult continuation and critical-wall arguments. These are useful local refinements and verification interfaces, not a full proof certification or a claim of a new solution of the Birkhoff conjecture.

A Exact Local Evidence and Reading Depth

The project root is `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`. The manifest row 283 associates the manuscript with commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. This is manifest provenance, not an independent verification of the checkout's Git object history. The exact catalogue row in `00-Applicability-Catalogue/applicability.json` and manifest row in `_sources/manuscripts.json` were read.

A.1 Pinned mathematical source

All S-files below are relative to the common directory

```
_sources/openai-math/preprints/Rigidity-of-Smooth-Billiards-with-a-Continu
ous-Caustic-Collar-September-24-2026/build
```

The complete `main.tex`, all eleven files below, the complete `references/references.bib`, and the five figure sources `figures/caustic.tex`, `figures/cone.tex`, `figures/monodromy.tex`, `figures/mesh.tex`, and `figures/last-site.tex` were read. Vendored style packages were not treated as mathematical evidence.

ID	File under `sections/`	Principal audit locators
S1	`01-introduction.tex`	`def:caustic-collar`; `thm:main`

ID	File under sections/	Principal audit locators
S2	02-analytic.tex	lem:rational-periodicity; thm:analytic; mesh construction, high rows, full Wiener inverse and Newton proof
S3	03-joint.tex	thm:joint-parametrization; rational value/jet compatibility, slanted domains and two difference divisions
S4	04-continuation.tex	cont:web-continuation; thm:lens-lines; thm:lens-points; lem:meromorphic-hartogs; thm:no-patch; cor:shape-natural-boundary
S5	05-shapes.tex	shape:isotropic-profile; shape:simple-centers; thm:isotropic-classification; thm:shape-lift; eq:V-coefficient
S6	06-cones.tex	lem:kernel-division; thm:cone; cone:no-quiet-witness; thm:finite-superlinear-exclusion
S7	07-profiles.tex	thm:exponential-obstruction; frozen inverse branches, corridor tracking; lem:log-deficit; thm:infinite-growth-exclusion
S8	08-critical.tex	thm:critical-bootstrap; prop:weighted-bootstrap; thm:no-trivial-profile; def:prohibited-sequence; thm:angle-budget-reduction
S9	09-model.tex	lem:contact-dichotomy; model:selected-fields; prop:pure-model; model:graph-filling; thm:mesh-contradiction; thm:sequence-exclusion
S10	10-odd-slice.tex	lem:odd-matrices; eq:cofactor; thm:boundary-contact; odd:modified-inverses; thm:stopped-geometry
S11	11-sign-test.tex	lem:discrete-tanaka; sign:large-moment; sign:witness-cost; thm:row-sign; prop:robust-sign; thm:last-site

A.2 Actual private corpus read

TR336, full mathematical report.
C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-336-astra.tex. All sections were read, including literal-target qualifications, eq:geometry, eq:residual, thm:exclusion, eq:rationalinverse, thm:dio, eq:fulltail, thm:action, thm:uniform, thm:block, thm:newton, and the final shortcut counterexamples. Its broader physical-to-phase gate is examined against S2's actual filled-collar argument rather than treated as a permanent obstruction.

TR250, selected exact foundation passages.
C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex. Read "What a Ram representation means in this report," eq:moments, and "A concrete polynomial Ram construction" (approximately lines 155–178 and 455–475). Only their finite unisolvent representation statement is used here.

TR247, finite complex-support theorem and proof.
C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR247-RH-AnotherResult/TR247-RH-another.tex. Read the entire finite moment-inertia section, including thm:moment-inertia, eq:moment-form, eq:moment-pencil, cor:finite-rh, and the following quotient-algebra setup (approximately lines 921–1081). Only finite moment/operator algebra with its node hypotheses is used; no global zeta conclusion is assumed.

TR2405, contour-count and Newton certificate.
`C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR168-Agentic-Program-Update/auto_reports_clv4/TR2405-auto-A-proof-of-Riemann-s-hypothesis-via-Hadamard-Wei/TR2405-revised-v2.tex`. Read "Certified Contour Inversion," `eq:count-contour`, `eq:moment-contour`, `eq:newton-contour`, and the contour recovery proposition, with the adjoining finite-product/global-remainder boundary (approximately lines 401–516). The interface requires full contour error and multiplicity accounting.

The maps `_audit/corpus-core-map.txt`, `_audit/corpus-foundations-map.txt`, and `_audit/corpus-applications-map.txt` were used for discovery and selected entries were read. They are not substitutes for the actual theorem passages just listed. No unlisted foundation theorem is claimed to have been checked through its full proof.

A.3 Reproduction boundary

The local helper `check_m283_refinements.py` accompanies this report. It checks finite algebraic identities and representative finite matrix calculations, not the global billiard theorem. The mathematical proofs, not the finite tests, establish the statements in this report. All original mathematical sources and corpus files are unchanged.

The complete independent source review `_audit/M283-source-audit.txt` was also read. Its conclusion is a bounded no-confirmed-defect result, not global certification. Its floating-point arbitrary-input Fourier checks were not rerun here and are not used as proofs. They check one real-axis input, selected modes and quadratures, not the full complex-strip operator bound. The refinements below were checked algebraically.

B Supplementary Analytic-Stage Refinements

B.1 A sharp Fourier-weight balance

The real-symmetric base-input hypothesis should be explicit when using S2, `prop:high-frequency-estimate`. Its relative strip loss uses imaginary displacement proportional to strip width; an unrestricted complex base input does not get that conclusion from an absolute displacement bound alone. This is a statement clarification for the source's intended real billiard inputs, not a confirmed defect in their application.

Suppose the source coefficient bound has the precise form

$$|R_{nk}| \le \frac{C_b}{|n|}\frac{e^{-(1-K/n^2)\sigma|n-k|}}{(1+|n-k|)^2}, \qquad |n| \ge N, \quad \sigma \ge 0.$$

Take $N^2 \ge 2K \ge 0$. For $n > 0$, put $\varepsilon = K/n^2$. The weight exponent is exactly

$$n - |k| - (1-\varepsilon)|n-k| = \begin{cases} \varepsilon(n-k), & 0 \le k \le n, \\ \varepsilon n - (2-\varepsilon)|k|, & k < 0, \\ -(2-\varepsilon)(k-n), & k > n. \end{cases}$$

It is at most K/n, attained at $k = 0$, improving the source's $2K/|n|$ estimate in `ana:exponential-balance`. Sign reversal treats negative n. For high inputs $|k| \ge N$, the opposite-sign case is nonpositive and the same-sign positive maximum is bounded by

$$\frac{K(n-N)}{n^2} \le \frac{K}{4N},$$

attained at $n = 2N, k = N$. Thus on Fourier–Wiener ℓ^1, with $S = \sum_{j\in\mathbb{Z}}(1+|j|)^{-2} = \pi^2/3 - 1$,

$$\|R\| \le \frac{C_b S}{N} e^{K\sigma/N}, \qquad \|R_H\| \le r_H := \frac{C_b S}{N} e^{K\sigma/(4N)}.$$

Indeed multiply each matrix entry by the output/input weight ratio and sum each column; $1/|n| \le 1/N$ and the remaining shift sum is at most S. If $r_H < 1$, the high-input/output operator $I + R_H$ has inverse norm at most $(1 - r_H)^{-1}$. The analogous coefficient-difference hypothesis retains its $1/N$ factor in the operator Lipschitz estimate. All these improvements are conditional on the nonlinear source row estimate; the case split does not prove that estimate.

B.2 Constant inputs beyond the normalized circle

For a real constant input $u = c$, write $A = e^c > 0$ and $b = 2\pi/q$. In the nondegenerate large-q regime, the stationary uniform mesh has variation kernel

$$K_A(t) = A^{-2}\left[2 + \cos(Ab/2) - 3\cos(A\min(t, b-t))\right], \quad 0 \le t \le b.$$

It follows directly by integrating the two half-cell kernels in S2, `ana:variation-kernel` at constant input. Symmetry of the cell and $qb = 2\pi$ give

$$\int_0^b K_A(t)e^{\mathrm{i}qt}\,\mathrm{d}t = -6A^{-2}\int_0^{\pi/q}\cos(At)\cos(qt)\,\mathrm{d}t = -\frac{6\sin(\pi A/q)}{A(q^2 - A^2)}.$$

Translation invariance makes the linearized Fourier rows diagonal; summing q cells and applying the source normalization gives

$$d_q(A) = \frac{q^3\sin(\pi A/q)}{\pi A(q^2 - A^2)} = \prod_{m=2}^{\infty}\left(1 - \frac{A^2}{q^2m^2}\right) \quad (q > A).$$

The product follows from the classical sine product after removing its first factor. It is positive, increases with q, and is less than one. For $q \ge 2A$, it is at least $8/(3\pi)$, obtained by substituting $A/q = 1/2$. On all high modes $|n| \ge N > A$, the diagonal inverse norm is exactly $1/d_N(A)$ in every exponential Fourier–ℓ^1 norm. Moreover,

$$d_q(A) = 1 - \left(\frac{\pi^2}{6} - 1\right)\frac{A^2}{q^2} + \left(1 - \frac{\pi^2}{6} + \frac{\pi^4}{120}\right)\frac{A^4}{q^4} + O(A^6/q^6).$$

This stronger $O(q^{-2})$ error is specific to constant inputs. It does not improve the general nonlinear $O(q^{-1})$ row estimate. A constant $A \ne 1$ is an arbitrary-input consistency test, not necessarily a closed billiard normalization. The scalar formula has a removable value at $q = A$, but this does not assert nondegenerate mesh uniqueness there.

B.3 A companion fixed-rational flux inverse

The companion report `M282-Analytic-Caustic-Collars/TR-M282-Ram-Collars.tex`, `thm:flux` and `cor:diskflux`, supplies a useful fixed-real-parameter improvement. Those theorem statements and proofs, and Section 5 of `_audit/M282-parent-derivations-audit.txt`, were read here; the entire companion report was not reaudited.

Fix reduced real $t = 2\pi\ell/q \ne 0$, let T be real translation by t, $\Pi = q^{-1}\sum_{j=0}^{q-1} T^j$, and $\partial = (T_{t/2} - T_{-t/2})/t$. On $\mathcal{K} = \ker\Pi$ in periodic supremum norm or a fixed bounded-holomorphic strip, write $R = \partial^{-1}$ and $\|R\| \le k_t = |t|\lfloor q^2/4\rfloor/q$. For bounded holomorphic multipliers $a, b = 1/a$, require $\inf|\Pi b| > 0$ throughout that domain. The inverse of $\partial a\partial$ on $\mathcal{K}$ is given by

$$H = Rf, \qquad h = -\frac{\Pi(bH)}{\Pi b}, \qquad v = R\big(b(H + h)\big).$$

Here h is a resonant *function*, not generally a scalar; $\Pi[b(H+h)] = 0$ is the exact flux compatibility. If $\|a - c\| \le \delta < c$ for a real constant $c > 0$ on the whole strip, the reciprocal-disk estimate gives

$$\left\|(\partial a\partial)^{-1}\right\| \le \frac{k_t^2 c}{(c-\delta)^2}; \qquad c = \tfrac{1}{4}: \quad \left\|(\partial a\partial)^{-1}\right\| \le \frac{4k_t^2}{(1-4\delta)^2}, \quad \delta < \tfrac{1}{4}.$$

Indeed b lies in the disk centered at $c/(c^2-\delta^2)$ with radius $\delta/(c^2-\delta^2)$, so $\|b\| \le 1/(c-\delta)$ and $\Re\Pi b \ge 1/(c+\delta)$. Applying these to the displayed flux formula yields the bound. The complete proof and the finite-q alternative bounds are in the companion report. The real translations are isometries; finite order by itself does not supply the norm estimate. Nothing here gives uniform complex-t jets, solves all rational resonance conditions, or establishes the global rigidity conclusion.

References

[1] A. Avila, J. De Simoi, and V. Kaloshin, An integrable deformation of an ellipse of small eccentricity is an ellipse, *Annals of Mathematics* 184 (2016), 527–558. https://doi.org/10.4007/annals.2016.184.2.5. Public publisher record checked.

[2] V. Kaloshin and A. Sorrentino, On the local Birkhoff conjecture for convex billiards, *Annals of Mathematics* 188 (2018), 315–380. https://doi.org/10.4007/annals.2018.188.1.6. Definition 6 and Main Theorem, p. 320, checked in the publisher PDF.

[3] M. Bialy and A. E. Mironov, The Birkhoff–Poritsky conjecture for centrally-symmetric billiard tables, *Annals of Mathematics* 196 (2022), 389–413. https://doi.org/10.4007/annals.2022.196.1.2. Publisher metadata and the theorem's scope as restated in the primary KKZ paper checked; the full proof was not reaudited.

[4] V. Kaloshin, C. E. Koudjinan, and K. Zhang, Birkhoff conjecture for nearly centrally symmetric domains, *Geometric and Functional Analysis* 34 (2024), 1973–2007. https://doi.org/10.1007/s00039-024-00695-6. Theorem 2 and Remark 3 checked.

[5] I. Koval, Local strong Birkhoff conjecture and local spectral uniqueness of almost every ellipse, *Inventiones Mathematicae* 244 (2026), 221–298; published online December 11, 2025. https://doi.org/10.1007/s00222-025-01397-y. Definition 1.3, Theorem 1, and publication data checked.

[6] A. A. Glutsyuk, On complex algebraic caustics in planar and projective billiards, *Regular and Chaotic Dynamics* 31 (2026), 132–149. https://doi.org/10.1134/S1560354726010016; https://arxiv.org/abs/2509.11257. Journal abstract and primary preprint record checked.

[7] C. Fierobe and A. Sorrentino, On the existence of periodic invariant curves for analytic families of twist maps and billiards, *Revista Matemática Iberoamericana*, https://doi.org/10.4171/RMI/1595. Publisher online-first PDF, copyright 2025, and https://arxiv.org/abs/2407.17090 checked for abstract scope. The pinned source lists 42 (2026), 299–322; that final issue pagination was not independently verified in this audit.

[8] *Rigidity of Smooth Billiards with a Continuous Caustic Collar*, September 24, 2026, pinned manuscript 283. Complete mathematical source and references read locally; exact files and labels are in Appendix A.

[9] M. Subbarao, private local Ram mathematical corpus: TR336; revised TR250; TR247, finite moment-inertia section; TR2405, contour inversion section. Exact editions, paths, locators, and reading boundaries are in Appendix A. These are local mathematical evidence, not public peer-review certification of the present source claim.